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3.2.1. Numpy: Matrix Operations

04:36

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()
3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition
print("Addition (A + B):")
print(matrix_a + matrix_b)

# Subtraction
print("Subtraction (A - B):")
print(matrix_a - matrix_b)

# Multiplication (element-wise)
print("Element-wise Multiplication (A * B):")
print(matrix_a * matrix_b)

# Matrix multiplication (dot product)
print("A dot B:")
print(np.dot(matrix_a, matrix_b))
```

```
# Transpose
```

```
print("Transpose of A:")
```

```
print(matrix_a.T)
```

Explorer

Average time
0.035 s
35.00 ms

Maximum time
0.057 s
57.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Debugger

Test case 1 57 ms Debug

Expected output

Actual output

Enter Matrix A:
1 2 3
4 5 6
7 8 9
Enter Matrix B:
1 1 1
1 1 1
1 1 1
Addition (A+B):
[[2 3 4]
[5 6 7]
[8 9 10]]
Subtraction (A-B):
[[0 1 2]
[3 4 5]
[6 7 8]]
Element-wise Multiplication (A*B):
[[1 2 3]
[4 5 6]
[7 8 9]]
A.dot.B:
[[6 6 6]
[15 15 15]
[24 24 24]]
Transpose of A:
[[1 4 7]
[2 5 8]
[3 6 9]]

Enter Matrix A:
1 2 3
4 5 6
7 8 9
Enter Matrix B:
1 1 1
1 1 1
1 1 1
Addition (A+B):
[[2 3 4]
[5 6 7]
[8 9 10]]
Subtraction (A-B):
[[0 1 2]
[3 4 5]
[6 7 8]]
Element-wise Multiplication (A*B):
[[1 2 3]
[4 5 6]
[7 8 9]]
A.dot.B:
[[6 6 6]
[15 15 15]
[24 24 24]]
Transpose of A:
[[1 4 7]
[2 5 8]
[3 6 9]]

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